

Project Title

Resume–Job Matching & Talent-Marketplace Engine

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

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ABSTRACT

The Resume–Job Matching & Talent Marketplace Engine is an intelligent recruitment platform that connects job seekers with employers through automated résumé analysis and smart job recommendations. The system leverages Artificial Intelligence, Natural Language Processing, and machine learning algorithms to extract relevant information such as skills, education, certifications, projects, and professional experience from resumes. It compares this information with job requirements to generate accurate matching scores and recommend suitable opportunities. Recruiters can efficiently filter, rank, and shortlist candidates based on predefined criteria, reducing manual effort and recruitment time. The platform also identifies skill gaps and suggests personalized learning resources to improve candidates' career prospects. By creating a centralized talent marketplace, the system enables seamless communication between employers and applicants while ensuring transparency and fairness in the hiring process. Overall, the proposed solution enhances recruitment efficiency, improves candidate-job compatibility, supports informed hiring decisions, and strengthens workforce development across diverse industries.

Signature of the Guide